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EXAMINER
TOTH, KAREN E

ART UNIT	PAPER NUMBER
3791	

NOTIFICATION DATE	DELIVERY MODE
12/29/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/283,371	Applicant(s) KRUSE et al.	
	Examiner KAREN E TOTH	Art Unit 3791	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 21 September 2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-18 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-18 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Information Disclosure Statement

2. The listing of references in the PCT international search report is not considered to be an information disclosure statement (IDS) complying with 37 CFR 1.98. 37 CFR 1.98(a)(2) requires a legible copy of: (1) each foreign patent; (2) each publication or that portion which caused it to be listed; (3) for each cited pending U.S. application, the application specification including claims, and any drawing of the application, or that portion of the application which caused it to be listed including any claims directed to that portion, unless the cited pending U.S. application is stored in the Image File Wrapper (IFW) system; and (4) all other information, or that portion which caused it to be listed. In addition, each IDS must include a list of all patents, publications, applications, or other information submitted for consideration by the Office (see 37 CFR 1.98(a)(1) and (b)), and MPEP § 609.04(a), subsection I. states, "the list ... must be submitted on a separate paper." Therefore, the references cited in the international search report have not been considered. Applicant is advised that the date of submission of any item of information in the international search report will be the date of submission of the IDS for purposes of determining compliance with the requirements for the IDS with 37 CFR 1.97, including all timing statement requirements of 37 CFR 1.97(e). See MPEP § 609.05(a).

Specification

3. The lengthy specification has not been checked to the extent necessary to determine the presence of all possible minor errors. Applicant's cooperation is requested in correcting any errors of which applicant may become aware in the specification.

4. Applicant is reminded of the proper language and format for an abstract of the disclosure.

The abstract should be in narrative form and generally limited to a single paragraph on a separate sheet within the range of 50 to 150 words in length. The abstract should describe the disclosure sufficiently to assist readers in deciding whether there is a need for consulting the full patent text for details.

The language should be clear and concise and should not repeat information given in the title. It should avoid using phrases which can be implied, such as, "The disclosure concerns," "The disclosure defined by this invention," "The disclosure describes," etc. In addition, the form and legal phraseology often used in patent claims, such as "means" and "said," should be avoided.

The abstract of the disclosure is objected to because it exceeds 150 words and uses "The invention relates to", which is a phrase that can be implied. Correction is required. See MPEP § 608.01(b).

Claim Interpretation

5. The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

6. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term “means” or “step” or a term used as a substitute for “means” that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term “means” or “step” or the generic placeholder is modified by functional language, typically, but not always linked by the transition word “for” (e.g., “means for”) or another linking word or phrase, such as “configured to” or “so that”; and

(C) the term “means” or “step” or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word “means” (or “step”) in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word “means” (or “step”) in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word “means” (or “step”) are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word “means” (or “step”) are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

7. This application includes one or more claim limitations that do not use the word “means,” but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with

functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not preceded by a structural modifier. Such claim limitation(s) is/are:

Sensor device, first recited in claim 16 (a polymer-optical fiber probe, p. 4 as filed)

Evaluation device, first recited in claim 16 (a computer or integrated circuit, p. 7 as filed)

Because this/these claim limitation(s) is/are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof.

If applicant does not intend to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation(s) recite(s) sufficient structure to perform the claimed function so as to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Objections

8. Claims 1-18 are objected to because of the following informalities:

The claims appear to be a literal translation into English from a foreign document and are replete with grammatical and idiomatic errors.

Claim 3 recites "at least two state transition models are provided, one on a constant glucose concentration"; unless the intent is to locate a model on a concentration somehow, it appears this should recite "**based** on a constant glucose concentration".

Claim 9 recites "determined as outliners"; it appears this should recite "determined as out**liers**".

Appropriate correction is required.

Claim Rejections - 35 USC § 112

9. The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

10. Claims 1-18 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

The term "several" in claim 2 is a relative term which renders the claim indefinite. The term "several" is not defined by the claim, the specification does not provide a standard for ascertaining the requisite degree, and one of ordinary skill in the art would not be reasonably apprised of the scope of the invention. It is not clear how many models are required to be

considered “several” or whether “several” might be more than a few, or any value more than one (that is, a plurality), or if it requires some specific number such as at least 5, at least 10, etc. Clarification is required.

Claim 4 recites “measured values determined are filtered” – it is unclear if the intent is that measured values are filtered? Or filtered measured values are determined? Or if both determined and filtered should be present in the claim? Clarification is required.

Claim 7 refers to “the null hypothesis”; there is insufficient antecedent basis for this limitation in the claim. It is unclear if this is some inherent aspect of the recited test or if this is merely a desired condition. Clarification is required. Further, this recitation of “the null hypothesis” is followed by an interjected aside of “the sample follows a mean-free Gaussian distribution with the determined variance of the measurement noise” – it is entirely unclear if this aside is considered to be the only possible null hypothesis? Or if this describes an example of a null hypothesis? Or if this describes an example of when a null hypothesis is not rejected? Further, it is unclear if this aside relates to any statistical test or only the optionally included Kolmogorov-Smirnov test. Clarification is required.

Claim 8 recites that variance of “at least one further sample” is performed “as long as the null hypothesis is rejected”. Whether the null hypothesis is rejected is only evaluated once as per claim 7, such that it is not clear if the intent is for claim 7 to be performed repeatedly? There does not appear to be any end condition if the null hypothesis is rejected that might result in it no longer being rejected. Clarification is required.

Claim 9 refers to “an NIS test” without providing the terms “NIS” is used to abbreviate; clarification is required.

Claim 10 calls for determining whether values are above or below a specified limit "before they are discarded"; claim 9, from which this depends, defines that "measured values that were determined as outliners [sic] are discarded" – is this check above/below a limit the check using at least one filter function? Or is this in addition to checking using the filter function? Clarification is required.

Claim 11 recites that "a current measured value" should be "rejected as a measurement error if at least a predetermined number… were previously rejected as measurement errors"; a predetermined number of measured values? Or a predetermined number of something else? It should be noted that the "in particular" clause is indefinite on its own merits and also only recited as an optional inclusion, such that the claim should be definite on its own merits. Further, even if the "in particular" clause is included, the claim as presented results in every measured value being rejected as a measurement value once two consecutive measurement values have occurred, since each subsequent value would also be rejected as a measurement value thus causing each further value beyond that to also be rejected as a measurement value. As presented the claim appears to be directed to an infinite loop of rejecting all values. Where there is a great deal of confusion and uncertainty as to the proper interpretation of the limitations of a claim, it would not be proper to reject such a claim on the basis of prior art. As stated in In re Steele, 305 F.2d 859, 134 USPQ 292 (CCPA 1962), a rejection should not be based on considerable speculation about the meaning of terms employed in a claim or assumptions that must be made as to the scope of the claims. See MPEP 2173.06. The claim is too indefinite to treat on the merits at this time.

Claim 13 defines that "several earlier measured values are filtered, in particular by means of a Kalman Fixed Interval Smoother". This is entirely indefinite – does this require the at least two measured value to include even more values such that some can be considered "earlier"? Or are these some other unrelated values? How and why are these "earlier" values obtained? What are these values representative of? Why are these values filtered and how do these "filtered" "earlier" values relate to any other part of the method? This step does not appear to relate to any other part of the method. Where there is a great deal of confusion and uncertainty as to the proper interpretation of the limitations of a claim, it would not be proper to reject such a claim on the basis of prior art. As stated in In re Steele, 305 F.2d 859, 134 USPQ 292 (CCPA 1962), a rejection should not be based on considerable speculation about the meaning of terms employed in a claim or assumptions that must be made as to the scope of the claims. See MPEP 2173.06. As there are multiple possible values that might be the intended "earlier measured values" and their filtering appears to be entirely unrelated to the remainder of the method, the claim is too indefinite to treat on the merits at this time, as is claim 14 which depends from it.

Claim 14 refers to "the Kalman Fixed Interval Smoother"; there is no antecedent basis for this limitation in the claim. It is unclear if this refers to the "at least one Kalman filter", the "at least one extended Kalman filter", or some other filter. Further, the claim refers to both "an RTS filter" and "an MBF filter" without defining the words that these are used to abbreviate. The scope of the claim is entirely unclear.

Claim 15 defines "the trend in blood sugar concentration is classified using a number of categories". There is insufficient antecedent basis for "the trend" in the claim. Further, it is

entirely unclear where or how this trend is obtained or determined as claim 1 only calls for finding a single glucose value and a trend inherently requires at least two values. Is “the trend” a trend of the measured values? Or is the intent to find a trend of estimated current glucose values? Where there is a great deal of confusion and uncertainty as to the proper interpretation of the limitations of a claim, it would not be proper to reject such a claim on the basis of prior art. As stated in In re Steele, 305 F.2d 859, 134 USPQ 292 (CCPA 1962), a rejection should not be based on considerable speculation about the meaning of terms employed in a claim or assumptions that must be made as to the scope of the claims. See MPEP 2173.06. As there are multiple possible values that might be the intended “trend”, the claim is too indefinite to treat on the merits at this time.

11. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 1 recites the broad recitation “a transport fluid”, and the claim also recites “in particular blood” which is the narrower statement of the limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

12. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting

claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 2 recites the broad recitation "depending on the course over time of the estimated current glucose value", and the claim also recites "in particular its rate of change over time" which is the narrower statement of the limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

13. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 4 recites the broad recitation "errors... of the sensor device", and the claim also recites "in particular measurement errors" which is the narrower statement of the limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

14. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 5 recites the broad recitation "the at least one measurement noise value is adjusted", and the claim also recites "in particular regularly" which is the narrower statement of the limitation. The claim(s) are considered indefinite

because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

15. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 6 recites the broad recitation "the variance of the measurement noise is determined", and the claim also recites "in particular it is estimated" which is the narrower statement of the limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

16. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 7 recites the broad recitation "a statistical test", and the claim also recites "in particular a Kolmogorov-Smirnov-test" which is the narrower statement of the limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

17. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 9 recites the broad recitation "using the at least one filter function", and the claim also recites "in particular using an NIS test" which is the narrower statement of the limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

18. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 13 recites the broad recitation "several earlier measurement values are filtered", and the claim also recites "in particular by means of a Kalman Fixed Interval Smoother" which is the narrower statement of the limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

19. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 15 recites the broad recitation "a number of

categories", and the claim also recites "in particular using at least seven categories" which is the narrower statement of the range/limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

20. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 16 recites the broad recitation "a transport fluid", and the claim also recites "in particular blood" which is the narrower statement of the limitation. Claim 16 further recites the broad recitation "a sensor device" and also recites "in particular for measuring fluorescence" which is the narrower statement of the limitation. Claim 16 further recites the broad recitation "a probe" and also recites "in particular a polymer-optical fiber probe" which is the narrower statement of the limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

21. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 17 recites the broad recitation "a transport fluid", and the claim also recites "in particular blood" which is the narrower statement of the

limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

22. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 18 recites the broad recitation “a transport fluid”, and the claim also recites “in particular blood” which is the narrower statement of the limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

23. Claim limitation “provision device”, first recited in claim 16, invokes 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. However, the written description fails to disclose the corresponding structure, material, or acts for performing the entire claimed function and to clearly link the structure, material, or acts to the function. No “provision device” has been identified, defined, or described in the disclosure. Therefore, the claim is indefinite and is rejected under 35 U.S.C. 112(b) or pre-AIA 35 U.S.C. 112, second paragraph.

Applicant may:

(a) Amend the claim so that the claim limitation will no longer be interpreted as a limitation under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph;

(b) Amend the written description of the specification such that it expressly recites what structure, material, or acts perform the entire claimed function, without introducing any new matter (35 U.S.C. 132(a)); or

(c) Amend the written description of the specification such that it clearly links the structure, material, or acts disclosed therein to the function recited in the claim, without introducing any new matter (35 U.S.C. 132(a)).

If applicant is of the opinion that the written description of the specification already implicitly or inherently discloses the corresponding structure, material, or acts and clearly links them to the function so that one of ordinary skill in the art would recognize what structure, material, or acts perform the claimed function, applicant should clarify the record by either:

(a) Amending the written description of the specification such that it expressly recites the corresponding structure, material, or acts for performing the claimed function and clearly links or associates the structure, material, or acts to the claimed function, without introducing any new matter (35 U.S.C. 132(a)); or

(b) Stating on the record what the corresponding structure, material, or acts, which are implicitly or inherently set forth in the written description of the specification, perform the claimed function. For more information, see 37 CFR 1.75(d) and MPEP §§ 608.01(o) and 2181.

Claim Rejections - 35 USC § 101

24. 35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

25. Claims 1-18 are rejected under 35 U.S.C. 101 because the claimed invention is directed to a judicial exception (i.e., a law of nature, a natural phenomenon, or an abstract idea) without significantly more. Claims 1-18 do not include additional elements that integrate the exception into a practical application of the exception or that are sufficient to amount to significantly more than the judicial exception for the reasons provided below which are in line with the 2014 Interim Guidance on Patent Subject Matter Eligibility (Federal Register, Vol. 79, No. 241, p 74618, December 16, 2014), the July 2015 Update on Subject Matter Eligibility (Federal Register, Vol. 80, No. 146, p. 45429, July 30, 2015), the May 2016 Subject Matter Eligibility Update (Federal Register, Vol. 81, No. 88, p. 27381, May 6, 2016), and the 2019 Revised Patent Subject Matter Eligibility Guidance (Federal Register, Vol. 84, No. 4, page 50, January 7, 2019).

<u>The analysis of claim 1 is as follows:</u>

Step 1: Claim 1 is drawn to a process, which is one of the four statutory categories.

Step 2A - Prong 1: Claim 1 is drawn to an abstract idea in the form of a process that, under its broadest reasonable interpretation, covers performance of the limitations in the mind but for the recitation of generic computer components. Additionally or alternatively, claim 1 is drawn to an abstract idea in the form of mathematical algorithms and/or formulas.

In particular, claim 1 recites the following limitations:

[A1]: determining the tissue glucose level using the series of measurements given, based on a sensor model, in which, by means of a sensor model, measurements of the sensor device are correlated to the tissue glucose levels while taking into account measurement noise,

[B1]: providing a state transition model, in which, by means of the state transition model, at least one glucose level in the transport fluid is correlated to the tissue glucose levels that have been determined while taking into account process noise, and

[C1]: estimating the current glucose level based on the state transition model that has been provided and the tissue glucose level that has been determined,

[D1]: in which at least one Kalman filter or at least one extended Kalman filter is used to provide the current glucose level;

These elements [A1]-[D1] of claim 1 are drawn to an abstract idea because they are processes that, under their broadest reasonable interpretation, are mere steps that are capable of being mentally performed with the aid of pen and paper. For example, a skilled artisan is capable of denoising tissue glucose measurements using a generic model in order to determine a tissue glucose level, convert the tissue glucose level to a glucose level of a transport fluid based on a generic model while taking into noise into account, and converting the transport fluid glucose level to a current glucose level based on a moving horizon estimation method . Additionally or alternatively, the elements [A1]-[D1] are drawn to an abstract idea because they are mathematical algorithms and/or formulas.

Step 2A - Prong Two: Claim 1 does not recite additional elements that integrate the judicial exception into a practical application. Claim 1 recites the following additional elements:

[A2]: determining, using a sensor device, a series of measurements comprising at least two measurements separated in time for a tissue glucose level in the tissue surrounding the transport fluid.

The element [A2] does not integrate the exception into a practical application of the exception because the element amounts to merely adding insignificant extra-solution activity to the judicial exception, e.g., mere data gathering at a higher level of generality in conjunction with the abstract idea that uses conventional, routine, and well known elements - see MPEP 2106.04(d); MPEP 2106.05(g). This extra-solution activity does not improve the functioning of technology, apply the judicial exception to effect a particular treatment, apply the judicial exception with a particular machine, or effect any sort of transformation.

Accordingly, the additional element does not integrate the abstract into a practical application because it does not impose any meaningful limitations on practicing the abstract idea.

Step 2B: Claim 1 does not recite additional elements that amount to significantly more than the judicial exception itself. Claim 1 recites the following additional elements:

[A2]: determining, using a sensor device, a series of measurements comprising at least two measurements separated in time for a tissue glucose level in the tissue surrounding the transport fluid

The element [A2] does not qualify as significantly more because the element amounts to merely adding insignificant extra-solution activity to the judicial exception, e.g., mere data gathering at a higher level of generality in conjunction with the abstract idea that uses

conventional, routine, and well known elements - see MPEP 2106.05(g). Additionally, the element is well-understood, routine, and conventional. US 5,408,999 A (Singh) teaches, in Col. 1, lines 41-57, fiber optic devices for measurement of blood gases, pH, electrolytes, and glucose are well known, and certain prior art sensors usually include an indicator molecule (dye) such as fluorescent or absorption dye which interacts with the component to be sensed or measured. The Examiner asserts that such fiber optic devices are capable of providing a series of measurements of tissue glucose levels.

In view of the above, the additional elements individually do not amount to significantly more than the above-judicial exception (the abstract idea). Looking at the limitations as an ordered combination (that is, as a whole) adds nothing that is not already present when looking at the elements taking individually. There is no indication that the combination of elements improves the functioning of a computer, for example, or improves any other technology. There is no indication that the combination of elements permits automation of specific tasks that previously could not be automated. There is no indication that the combination of elements includes a particular solution to a computer-based problem or a particular way to achieve a desired computer-based outcome. Rather, the collective functions of the claimed invention merely provide conventional computer implementation, i.e., the computer is simply a tool to perform the process.

Independent claim 17 recites mirrored device and storage medium limitations and is not patent eligible for substantially similar reasons.

Claim 16 recites the additional element of a sensor device, particularly for measuring fluorescence in a tissue surrounding the transport fluid, by means of a fiber optic probe,

designed to determine a series of measurements, comprising at least two measurements separated in time for a tissue surrounding the transport fluid. The element does not amount to a practical application or qualify as significantly more because the element amounts to merely adding insignificant extra-solution activity to the judicial exception, e.g., mere data gathering at a higher level of generality in conjunction with the abstract idea that uses conventional, routine, and well known elements - see MPEP 2106.04(d); MPEP 2106.05(g). Additionally, the element is well-understood, routine, and conventional. US 5,408,999 A (Singh) teaches, in Col. 1, lines 41-57, fiber optic devices for measurement of blood gases, pH, electrolytes, and glucose are well known, and certain prior art sensors usually include an indicator molecule (dye) such as fluorescent or absorption dye which interacts with the component to be sensed or measured. The Examiner asserts that such fiber optic devices are capable of providing a series of measurements of tissue glucose levels.

Claim 17 recites the additional elements of “an interface to connect a sensor device”, “at least one memory”, and “a calculating device”. These elements do not amount to a practical application or qualify as significantly more because these elements are simply appending well-understood, routine and conventional activities previously known in the industry, specified at a high level of generality, to the judicial exception, e.g., a claim to an abstract idea requiring no more than a generic computer to perform generic computer functions that are well-understood, routine and conventional activities previously known in the industry (See MPEP 2106.05(d)(II); Symantec, 838 F.3d at 1321, 120 USPQ2d at 1362 (utilizing an intermediary computer to forward information); TLI Communications LLC v. AV Auto. LLC, 823 F.3d 607, 610, 118 USPQ2d 1744, 1745 (Fed. Cir. 2016) (using a telephone for image transmission); OIP Techs.,

Inc., v. Amazon.com, Inc., 788 F.3d 1359, 1363, 115 USPQ2d 1090, 1093 (Fed. Cir. 2015) (sending messages over a network); buySAFE, Inc. v. Google, Inc., 765 F.3d 1350, 1355, 112 USPQ2d 1093, 1096 (Fed. Cir. 2014) (computer receives and sends information over a network)) and/or a claim to an abstract idea requiring no more than being stored on a computer readable medium which is a well-understood, routine and conventional activity previously known in the industry (See MPEP 2106.05(d)(II); Versata Dev. Group, Inc. v. SAP Am., Inc., 793 F.3d 1306, 1334, 115 USPQ2d 1681, 1701 (Fed. Cir. 2015); OIP Techs., 788 F.3d at 1363, 115 USPQ2d at 1092-93).

Claims 2-16 and 18 depend from claim 1, and they recite the same abstract idea as claim 1. Furthermore, these claims only contain recitations that further limit the abstract idea (that is, the claims only recite limitations that further limit the mental process or mathematical algorithm) and/or append abstract ideas (that is, the claims only recite limitations that add further mental processes or mathematical algorithms).

Claim 16 recites the additional element of a sensor device, particularly for measuring fluorescence in a tissue surrounding the transport fluid, by means of a fiber optic probe, designed to determine a series of measurements, comprising at least two measurements separated in time for a tissue surrounding the transport fluid. The element does not amount to a practical application or qualify as significantly more because the element amounts to merely adding insignificant extra-solution activity to the judicial exception, e.g., mere data gathering at a higher level of generality in conjunction with the abstract idea that uses conventional, routine, and well known elements - see MPEP 2106.04(d); MPEP 2106.05(g). Additionally, the element is well-understood, routine, and conventional. US 5,408,999 A (Singh) teaches, in Col.

1, lines 41-57, fiber optic devices for measurement of blood gases, pH, electrolytes, and glucose are well known, and certain prior art sensors usually include an indicator molecule (dye) such as fluorescent or absorption dye which interacts with the component to be sensed or measured. The Examiner asserts that such fiber optic devices are capable of providing a series of measurements of tissue glucose levels.

Claim 16 recites the additional elements of "a provision device" and "an evaluating device". These elements do not amount to a practical application or qualify as significantly more because these elements are simply appending well-understood, routine and conventional activities previously known in the industry, specified at a high level of generality, to the judicial exception, e.g., a claim to an abstract idea requiring no more than a generic computer to perform generic computer functions that are well-understood, routine and conventional activities previously known in the industry (See MPEP 2106.05(d)(II); Symantec, 838 F.3d at 1321, 120 USPQ2d at 1362 (utilizing an intermediary computer to forward information); TLI Communications LLC v. AV Auto. LLC, 823 F.3d 607, 610, 118 USPQ2d 1744, 1745 (Fed. Cir. 2016) (using a telephone for image transmission); OIP Techs., Inc., v. Amazon.com, Inc., 788 F.3d 1359, 1363, 115 USPQ2d 1090, 1093 (Fed. Cir. 2015) (sending messages over a network); buySAFE, Inc. v. Google, Inc., 765 F.3d 1350, 1355, 112 USPQ2d 1093, 1096 (Fed. Cir. 2014) (computer receives and sends information over a network)) and/or a claim to an abstract idea requiring no more than being stored on a computer readable medium which is a well-understood, routine and conventional activity previously known in the industry (See MPEP 2106.05(d)(II); Versata Dev. Group, Inc. v. SAP Am., Inc., 793 F.3d 1306, 1334, 115 USPQ2d 1681, 1701 (Fed. Cir. 2015); OIP Techs., 788 F.3d at 1363, 115 USPQ2d at 1092-93).

In view of the above, the additional elements do not integrate the abstract idea into a practical application and do not amount to significantly more than the above-judicial exception (the abstract idea). Looking at the limitations as an ordered combination (that is, as a whole) adds nothing that is not already present when looking at the elements taking individually. There is no indication that the combination of elements improves the functioning of a computer, for example, or improves any other technology. There is no indication that the combination of elements permits automation of specific tasks that previously could not be automated. There is no indication that the combination of elements includes a particular solution to a computer-based problem or a particular way to achieve a desired computer-based outcome. Rather, the collective functions of the claimed invention merely provide conventional computer implementation, i.e., the computer is simply a tool to perform the process.

Claim Rejections - 35 USC § 102

26. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

27. Claim(s) 1, 4, 5, 9, 10, 12, 16-18 is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Krause (WO 2020211910; citations below are to the corresponding US version US 2022/0192544).

Regarding claim 1, Kruse discloses a method for, in particular, continuously determining a current glucose value in a transport fluid, in particular blood, of an organism, comprising the steps of

a) Determining (S1) a measurement series using a sensor device, comprising at least two measured values for a tissue glucose value that are spaced apart in time in the tissue surrounding the transport fluid (paragraph [0013]),

b) determining (S2) the tissue glucose value using the determined series of measurements based on a measurement model in the form of a linear or non-linear function, with the measurement model measuring values of the sensor device measuring tissue glucose values taking into account at least one measurement noise value are assigned (paragraph [0014]),

c) providing (S3) at least one state transition model, with the at least one state transition model being used to assign at least one glucose value in the transport fluid to the determined tissue glucose values, taking into account at least one process noise value (paragraph [0015]), and

d) estimating (S4) the current glucose value in the transport fluid based on an approximation of at least one provided state transition model and the determined tissue glucose value using at least one Kalman filter in the case of a measurement model in the form

of a linear function or at least one extended Kalman filter Case of a measurement model in the form of a non-linear function (paragraphs [0010], [0016], [0107]).

Regarding claim 4, Krause further discloses that measured values determined are filtered by means of at least one filter function, errors, in particular measurement errors, of the sensor device being suppressed by means of the at least one filter function (paragraph [0042]).

Regarding claim 5, Krause further discloses that the at least one measurement noise value is adjusted, in particular regularly (paragraphs [0037]-[0038]).

Regarding claim 9, Krause further discloses that that measured values are checked for outliners using the at least one filter function and measured values that were determined as outliners are discarded (paragraph [0045]).

Regarding claim 10, Krause further discloses that the measured values are checked using the at least one filter function to determine whether they are above or below specified limit values before they are discarded (paragraph [0045]).

Regarding claim 12, Krause further discloses that the state transition model comprises a diffusion model for time-dependent modelling of the diffusion process of glucose from the transport fluid into the surrounding tissue (paragraph [0047]).

Regarding claim 16, Krause further discloses a device for in particular continuously determining a current glucose value in a transport fluid, in particular blood, of an organism, in particular for carrying out a method according to claim 1,

comprising a sensor device, in particular for measuring fluorescence in the transport fluid surrounding tissue by means of a probe, designed to determine a series of measurements,

comprising at least two measured values spaced apart in time for a tissue glucose value in the tissue surrounding the transport fluid (paragraph [0017],

a provision device, designed to provide at least one state transition model, with the at least one state transition model being used to assign at least one glucose value in the transport fluid to the determined tissue glucose values, taking into account at least one process noise value, and to provide a measurement model in the form of a linear or non-linear function, whereby measured values of the sensor device are assigned to tissue glucose values by means of the measurement model, taking into account at least one measurement noise value (paragraph [0017]), and

an evaluation device, designed to determine the tissue glucose value using the determined series of measurements based on the measurement model and to estimate the current glucose value in the transport fluid based on an approximation of at least one provided state transition model and the determined tissue glucose value using at least one Kalman filter in the case of a measurement model in form of a linear function or at least an extended Kalman filter in the case of a measurement model in the form of a non-linear function (paragraphs [0010], [0018], [0107]).

Regarding claim 17, Krause discloses an evaluation device for, in particular, continuously determining a current glucose value in a transport fluid, in particular blood, of an organism, comprising

at least one interface for connecting a sensor device for providing a series of measurements, comprising at least two measured values at different times for a tissue glucose value in the tissue surrounding the transport fluid (paragraph [0019]),

at least one memory for storing at least one state transition model, wherein the at least one state transition model is used to assign the tissue glucose values determined by the at least one state transition model to at least one glucose value in the transport fluid to, taking into account at least one process noise value and for storing a measurement model in the form of a linear or non- linear function, with the measurement model being used to assign measured values of the sensor device to tissue glucose values, taking into account at least one measurement noise value (paragraph [0019]), and

a computing device, designed to determine the tissue glucose value using the determined series of measurements based on the stored measurement model and for estimating the current glucose value in the transport fluid based on an approximation of at least one provided state transition model and the determined tissue glucose value using at least one Kalman filter in the case of a measurement model in the form of a linear function or at least one extended Kalman filter in the case of a measurement model in the form of a non-linear function (paragraphs [0010], [0020], [0107]).

Regarding claim 18, Krause further discloses non-transitory, computer-readable medium for storing instructions (paragraph [0021]) which, executed on a computer, cause a method for, in particular, continuous determination of a current glucose value in a transport fluid, in particular blood, of an organism to be carried out, preferably suitable for carrying out a method of claim 1, comprising the steps of

a) Determining by means of a sensor device a series of measurements comprising at least two time-spaced measured values for a tissue glucose value in the tissue surrounding the transport fluid (paragraph [0022]),

b) determining the tissue glucose value using the determined series of measurements based on a measurement model in the form of a linear or non-linear function, with the measurement model being used to assign measured values of the sensor device to tissue glucose values, taking into account at least one measurement noise value (paragraph [0023]),

c) providing at least one state transition model, with the at least one state transition model using the determined tissue glucose at least one glucose value in the transport fluid is assigned to glucose values, taking into account at least one process noise value (paragraph [0024]), and

d) estimating the current glucose value in the transport fluid based on an approximation of at least one provided state transition model and the determined tissue glucose value using at least one Kalman filter in the case of a measurement model in the form of a linear function or at least one extended Kalman filter in the case of a measurement model in the form of a non-linear function (paragraphs [0010], [0025], [0107]).

28. Claim(s) 1, 4-10, 12, 17, 18 is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Singh (US 2020/0383643).

Regarding claim 1, Singh discloses a method for, in particular, continuously determining a current glucose value in a transport fluid, in particular blood, of an organism (paragraphs [0022], [0025]), comprising the steps of

a) Determining (S1) a measurement series using a sensor device, comprising at least two measured values for a tissue glucose value that are spaced apart in time in the tissue surrounding the transport fluid (paragraph [0025]; figures 4, 6);

b) determining (S2) the tissue glucose value using the determined series of measurements based on a measurement model in the form of a linear or non-linear function (paragraphs [0087], [0121], [0122]), with the measurement model measuring values of the sensor device measuring tissue glucose values taking into account at least one measurement noise value are assigned (paragraphs [0087], [0123]),

c) providing (S3) at least one state transition model, with the at least one state transition model being used to assign at least one glucose value in the transport fluid to the determined tissue glucose values (paragraphs [0082], [0083], [0112]), taking into account at least one process noise value (paragraph [0096], [0104]), and

d) estimating (S4) the current glucose value in the transport fluid based on an approximation of at least one provided state transition model and the determined tissue glucose value using at least one Kalman filter in the case of a measurement model in the form of a linear function or at least one extended Kalman filter Case of a measurement model in the form of a non-linear function (paragraphs [0012], [0082], [0083], [0087], [0105], [0121]).

Regarding claim 4, Singh further discloses that measured values determined are filtered by means of at least one filter function, errors, in particular measurement errors, of the sensor device being suppressed by means of the at least one filter function (paragraph [0071]).

Regarding claim 5, Singh further discloses that the at least one measurement noise value is adjusted, in particular regularly (paragraph [0071]).

Regarding claim 6, Singh further discloses that, in order to adapt the at least one measurement noise value, the variance of the measurement noise is determined using a random sample of measured values, in particular it is estimated (paragraph [0117]).

Regarding claim 7, Singh further discloses that a statistical test is used to check whether a null hypothesis is not rejected (paragraph [0071]).

Regarding claim 8, Singh further discloses that the variance of the measurement noise is determined for at least one further sample of measured values as long as the null hypothesis is rejected (paragraphs [0071], [0072]).

Regarding claim 9, Singh further discloses that measured values are checked for outliners using the at least one filter function and measured values that were determined as outliners are discarded (paragraph [0071].

Regarding claim 10, Singh further discloses that the measured values are checked using the at least one filter function to determine whether they are above or below specified limit values before they are discarded (paragraph [0071]).

Regarding claim 12, Singh further discloses that the state transition model comprises a diffusion model for time-dependent modelling of the diffusion process of glucose from the transport fluid into the surrounding tissue.

Regarding claim 17, Singh discloses an evaluation device for, in particular, continuously determining a current glucose value in a transport fluid, in particular blood, of an organism (paragraph [0030]), comprising

at least one interface for connecting a sensor device for providing a series of measurements (paragraph [0043], [0046]), comprising at least two measured values at different times for a tissue glucose value in the tissue surrounding the transport fluid (paragraph [0025; figures 4, 6),

at least one memory (paragraph [0047]) for storing at least one state transition model, wherein the at least one state transition model is used to assign the tissue glucose values determined by the at least one state transition model to at least one glucose value in the transport fluid to (paragraphs [0082], [0083], [0112]), taking into account at least one process noise value (paragraphs [0096], [0104]) and for storing a measurement model in the form of a linear or non- linear function, with the measurement model being used to assign measured values of the sensor device to tissue glucose values (paragraphs [0087], [0121], [0122]), taking into account at least one measurement noise value (paragraphs [0087], [0123]), and

a computing device (paragraphs [0043]-[0047]), designed to determine the tissue glucose value using the determined series of measurements based on the stored measurement model and for estimating the current glucose value in the transport fluid based on an approximation of at least one provided state transition model and the determined tissue glucose value using at least one Kalman filter in the case of a measurement model in the form of a linear function or at least one extended Kalman filter in the case of a measurement model in the form of a non-linear function (paragraphs [0012], [0082], [0083], [0087], [0105], [0121]).

Regarding claim 18, Singh discloses a non-transitory, computer-readable medium for storing instructions (paragraph [0030]) which, executed on a computer, cause a method for, in particular, continuous determination of a current glucose value in a transport fluid, in particular

blood, of an organism to be carried out, preferably suitable for carrying out a method according to claim 1 (paragraph [0031], [0032]), comprising the steps of a) Determining (S1) a measurement series using a sensor device, comprising at least two measured values for a tissue glucose value that are spaced apart in time in the tissue surrounding the transport fluid (paragraph [0025]; figures 4, 6);

b) determining (S2) the tissue glucose value using the determined series of measurements based on a measurement model in the form of a linear or non-linear function (paragraphs [0087], [0121], [0122]), with the measurement model measuring values of the sensor device measuring tissue glucose values taking into account at least one measurement noise value are assigned (paragraphs [0087], [0123]),

c) providing (S3) at least one state transition model, with the at least one state transition model being used to assign at least one glucose value in the transport fluid to the determined tissue glucose values (paragraphs [0082], [0083], [0112]), taking into account at least one process noise value (paragraph [0096], [0104]), and

d) estimating (S4) the current glucose value in the transport fluid based on an approximation of at least one provided state transition model and the determined tissue glucose value using at least one Kalman filter in the case of a measurement model in the form of a linear function or at least one extended Kalman filter Case of a measurement model in the form of a non-linear function (paragraphs [0012], [0082], [0083], [0087], [0105], [0121]).

Claim Rejections - 35 USC § 103

29. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

30. Claim(s) 2 and 3 is/are rejected under 35 U.S.C. 103 as being unpatentable over Krause in view of Barceló Rico (Barceló Rico, F. (2012). Multimodel Approaches for Plasma Glucose Estimation in Continuous Glucose Monitoring. Development of New Calibration Algorithms [Tesis doctoral]. Universitat Politècnica de València. https://doi.org/10.4995/Thesis/10251/17173).

Regarding claims 2 and 3, Krause does not disclose that several state transition models are provided, which are changed depending on the course over time of the estimated current glucose value, in particular its rate of change over time, the “several” models including one based on a constant glucose concentration, one based on a constant change in glucose concentration and/or one based on a weighted sum of previous glucose concentrations. Barceló Rico teaches a method of estimating a current glucose value in a transport fluid which comprises providing several state transition models which are dependent upon a current value and which are based on a constant concentration, a changing concentration, and/or a weighted sum of concentrations (p. 26, lines 10-18; p. 42, lines 2-12; p. 75, lines 6-7). It would have been obvious to one of ordinary skill in the art at the time the invention was filed to have followed

Krause and included a plurality of state transition models, as taught by Barceló Rico, in order to improve the accuracy of estimation.

31. Claim(s) 2 and 3 is/are rejected under 35 U.S.C. 103 as being unpatentable over Singh in view of Barceló Rico (Barceló Rico, F. (2012). Multimodel Approaches for Plasma Glucose Estimation in Continuous Glucose Monitoring. Development of New Calibration Algorithms [Tesis doctoral]. Universitat Politècnica de València. https://doi.org/10.4995/Thesis/10251/17173).

Regarding claims 2 and 3, Singh does not disclose that several state transition models are provided, which are changed depending on the course over time of the estimated current glucose value, in particular its rate of change over time, the “several” models including one based on a constant glucose concentration, one based on a constant change in glucose concentration and/or one based on a weighted sum of previous glucose concentrations. Barceló Rico teaches a method of estimating a current glucose value in a transport fluid which comprises providing several state transition models which are dependent upon a current value and which are based on a constant concentration, a changing concentration, and/or a weighted sum of concentrations (p. 26, lines 10-18; p. 42, lines 2-12; p. 75, lines 6-7). It would have been obvious to one of ordinary skill in the art at the time the invention was filed to have followed Krause and included a plurality of state transition models, as taught by Barceló Rico, in order to improve the accuracy of estimation.

32. Claim(s) 16 is/are rejected under 35 U.S.C. 103 as being unpatentable over Singh in view of Sia (US 2018/0177439).

Singh further discloses a device for in particular continuously determining a current glucose value in a transport fluid, in particular blood, of an organism, in particular for carrying out a method according to claim 1, comprising

a sensor device, in particular for measuring a characteristic in the transport fluid surrounding tissue to determine a series of measurements, comprising at least two measured values spaced apart in time for a tissue glucose value in the tissue surrounding the transport fluid (paragraph [0025], figures 4, 6),

a provision device (paragraphs [0046], [0047]) , designed to provide at least one state transition model, with the at least one state transition model being used to assign at least one glucose value in the transport fluid to the determined tissue glucose values (paragraphs [0082], [0083], [0112]), taking into account at least one process noise value (paragraphs [0096], [0104]), and to provide a measurement model in the form of a linear or non-linear function, whereby measured values of the sensor device are assigned to tissue glucose values by means of the measurement model (paragraphs [0087], [0121], [0122]), taking into account at least one measurement noise value (paragraphs [0087], [0123]), and

an evaluation device (paragraphs [0043]-[0047]), designed to determine the tissue glucose value using the determined series of measurements based on the measurement model and to estimate the current glucose value in the transport fluid based on an approximation of at least one provided state transition model and the determined tissue glucose value using at least one Kalman filter in the case of a measurement model in form of a linear function or at

least an extended Kalman filter in the case of a measurement model in the form of a non-linear function (paragraphs [0012], [0082], [0083], [0087], [0105], [0121]).

Singh does not disclose the sensor device being configured for measuring fluorescence in the transport fluid surrounding tissue by means of a probe, in particular a polymer-optical fiber probe. Sia teaches a method of continuously determining a current glucose value in a transport fluid comprising determining a measurement series of at least two measured values using a polymer-optical fiber probe (paragraphs [0042], [0062], [0068]) that is used to estimate a current glucose value (paragraphs [0071], [0073]). It would have been obvious to one of ordinary skill in the art at the time the invention was filed to have followed Singh using a polymer-optical fiber probe to determine the measurement series, as taught by Sia, in order to "limit tissue damage" and allow "painless" observation (paragraph [0073] of Sia).

Conclusion

33. Any inquiry concerning this communication or earlier communications from the examiner should be directed to KAREN E TOTH whose telephone number is (571)272-6824. The examiner can normally be reached Mon - Fri 9a-6p.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jennifer Robertson can be reached at 571-272-5001. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/KAREN E TOTH/
Examiner, Art Unit 3791